

Chapter 9

Squeeze Theorem for Sequences. Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ with $a_n \leq b_n \leq c_n$ for all integers n greater than some index N . If $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

Growth Rate of Sequences. The following sequences are ordered according to increasing growth rates as $n \rightarrow \infty$: $\{\ln n\} \ll \{n^p\} \ll \{n^p \ln n\} \ll \{n^{p+s}\} \ll \{b^n\} \ll \{n!\} \ll \{n^n\}$. The ordering applies for positive real numbers p, r , and s and $b > 1$.

Geometric Series. Let $r, a \in \mathbb{R}$. If $|r| < 1$, then $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$. If $|r| \geq 1$, then the series diverges.

Divergence Test. If $\sum a_k$ converges, then $\lim_{n \rightarrow \infty} a_k = 0$. Equivalently, if $\lim_{n \rightarrow \infty} a_k \neq 0$, then the series diverges.

Integral Test. Suppose f is a continuous, positive, decreasing function for $x \geq 1$ and let $a_k = f(k)$ for $k = 1, 2, 3, \dots$. Then

$$\sum_{k=1}^{\infty} a_k \text{ and } \int_1^{\infty} f(x) dx$$

either both converge or both diverge. In the case of convergence, the value of the integral is *not*, in general, equal to the value of the series.

Ratio Test. Let $\sum a_k$ be an infinite series with positive terms and let $r = \lim_{n \rightarrow \infty} \frac{a_{k+1}}{a_k}$.

1. If $0 \leq r < 1$ then the series converges.
2. If $r > 1$ (including $r = \infty$) then the series diverges.
3. If $r = 1$ then the test is inconclusive.

Root Test. Let $\sum a_k$ be an infinite series with positive terms and let $\rho = \lim_{n \rightarrow \infty} \sqrt[n]{a_k}$.

1. If $0 \leq \rho < 1$ then the series converges.
2. If $\rho > 1$ (including $r = \infty$) then the series diverges.
3. If $\rho = 1$ then the test is inconclusive.

Comparison Test. Let $\sum a_k$ and $\sum b_k$ be series with positive terms.

1. If $0 < a_k \leq b_k$ and $\sum b_k$ converges, then $\sum a_k$ converges.
2. If $0 < b_k \leq a_k$ and $\sum b_k$ diverges, then $\sum a_k$ diverges.

The Limit Comparison Test. Let $\sum a_k$ and $\sum b_k$ be series with positive terms and let $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = L$.

1. If $0 < L < \infty$ then $\sum a_k$ and $\sum b_k$ either both converge or diverge.
2. If $L = 0$ and $\sum b_k$ converges, then $\sum a_k$ converges.
3. If $L = \infty$ and $\sum b_k$ diverges, then $\sum a_k$ diverges.

Chapter 10

Taylor's Polynomials. Let f be a function with $f', f'', \dots, f^{(n)}$ defined at a . The **n th-order Taylor Polynomial** for f with its **center** at a denoted p_n , has the property that it matches f in value, slope, and all derivatives up to the n th derivative at a ; that is,

$$p_n(a) = f(a) \text{ and } p'_n(a) = f'(a), \dots, p_n^{(n)}(a) = f^{(n)}(a).$$

The n th-order Taylor polynomial centered at a is

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

More compactly,

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k \text{ for } k = 0, 1, 2, \dots, n.$$

Chapter 11

Slope of a Tangent Line. Let f be differentiable function at θ_0 . The slope of the line tangent to the curve $r = f(\theta)$ at the point $(f(\theta), \theta_0)$ is

$$\frac{dy}{dx} = \frac{f'(\theta_0) \sin \theta_0 + f(\theta_0) \cos \theta_0}{f'(\theta_0) \cos \theta_0 - f(\theta_0) \sin \theta_0},$$

provided the denominator is nonzero at the point. At angles θ_0 for which $f(\theta_0) = 0$ and $f'(\theta_0) \neq 0$, the tangent line is $\theta = \theta_0$ with slope $\tan \theta_0$.

Area of Regions in Polar Coordinates. Let R be the region bounded by the graph of $r = f(\theta)$ and the origin, between $\theta = \alpha$ and $\theta = \beta$, where f is continuous and $f(\theta) \geq g(\theta) \geq 0$ on $[\alpha, \beta]$. The area of R is

$$\int_{\alpha}^{\beta} \frac{1}{2} f(\theta)^2 d\theta.$$

12.3 Dot Products

Dot Product. Given two vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$,

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v} &= u_1 v_1 + u_2 v_2 + u_3 v_3 \\ &= |\mathbf{u}| |\mathbf{v}| \cos \theta\end{aligned}$$

Orthogonal Projection of $\mathbf{u}$ onto $\mathbf{v}$,

$$\begin{aligned}\text{proj}_{\mathbf{v}} \mathbf{u} &= |\mathbf{u}| \cos \theta \left(\frac{\mathbf{v}}{|\mathbf{v}|} \right) \\ &= \text{scal}_{\mathbf{v}} \mathbf{u} \left(\frac{\mathbf{v}}{|\mathbf{v}|} \right) \\ &= \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \right) \mathbf{v}.\end{aligned}$$

Scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$,

$$\text{scal}_{\mathbf{v}} \mathbf{u} = |\mathbf{u}| \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}.$$

Work. Let $\mathbf{F}$ be a constant force applied to object, producing a displacement $\mathbf{d}$. If the angle between $\mathbf{F}$ and $\mathbf{d}$ is θ , then the **work** done by the force is

$$W = |\mathbf{F}| |\mathbf{d}| \cos \theta = \mathbf{F} \cdot \mathbf{d}.$$

12.4 Cross Products

Cross product, of two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$,

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta$$

$$|\tau| = |\mathbf{r}| |\mathbf{F}| \sin \theta$$

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k}.$$

13 Functions of Several Variables

Chain Rule (Two Independent Variables). Let z be a differentiable function of x and y , where x and y are differentiable functions of s and t . Then

$$\frac{dz}{dt} = \frac{dz}{dx} \frac{dx}{dt} + \frac{dz}{dy} \frac{dy}{dt}.$$

Gradient (Two Dimensions). Let f be differentiable at the point (x, y) . The **gradient** of f at (x, y) is the vector-valued function

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}.$$

Directional Derivative. Let f be differentiable at (a, b) and let $\mathbf{u} = \langle u_1, u_2 \rangle$ be a unit vector in the xy -plane, the **directional derivative of f at (a, b) in the direction of $\mathbf{u}$** is

$$D_{\mathbf{u}}f(a, b) = \langle f_x(a, b), f_y(a, b) \rangle \cdot \langle u_1, u_2 \rangle$$

$$D_{\mathbf{u}}f(a, b) = \nabla f(a, b) \cdot \mathbf{u}.$$

Second Derivative Test. Suppose that the second partial derivatives of f are continuous throughout an open disk centered at the point (a, b) , where $f_x(a, b) = f_y(a, b) = 0$. Let $D(x, y) = f_{xx}f_{yy} - f_{xy}^2$.

1. If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum value at (a, b) .
2. If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum value at (a, b) .
3. If $D(a, b) < 0$, then f has a saddle point at (a, b) .
4. If $D(a, b) = 0$, then the test is inconclusive.

Helpful formulas

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$ (Chain rule)
3. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
4. $\frac{d}{dx}e^x = e^x$
5. $\frac{d}{dx}\sin ax = a \cos ax$
6. $\frac{d}{dx}\cos ax = -a \sin ax$
7. $\int x^n dx = \frac{1}{n+1}x^{n+1} + C; n \neq -1$
8. $\int \cos ax dx = \frac{1}{a}\sin ax + C$
9. $\int \sin ax dx = -\frac{1}{a}\cos ax + C$
10. $\int \cos^2 x dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$
11. $\int \sin^2 x dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$

